Customer Segmentation & Churn Pattern Analytics in European Banking

Unified Mentor Data Analyst Internship Project Analysis date: 14 October 2026 **Source file:** European_Bank.csv

Abstract

This project analyzes 10,000 European banking customer records to understand churn patterns across geography, demographics, tenure, activity, credit score, products, and account balance. The overall churn rate is **20.4%**. Germany has the highest country-level churn rate at **32.4%**, while customers aged 46-59 have the highest age-group churn rate at **51.1%**. Inactive customers churn at **26.9%**, compared with **14.3%** for active customers. The analysis also identifies high-balance customers using a transparent top-quartile proxy and measures potential balance exposure without incorrectly treating it as confirmed revenue loss.

The deliverables are a reproducible analysis, an interactive Streamlit dashboard, this research paper, and an executive summary. Findings describe associations in a customer snapshot; they do not establish causality.

1. Background and business context

Customer churn creates three business pressures for retail banks:

- lower future customer lifetime value;
- higher replacement and acquisition costs; and
- less stable revenue and relationship planning.

The project brief asks for segmentation-driven analysis rather than one overall churn percentage. The business questions are:

1. Which customer groups are most likely to churn?
2. How does churn differ across countries, age groups, and financial profiles?
3. Is churn concentrated among high-value or low-value customers?

2. Problem statement and objectives

Banks may have customer-level data but still lack practical answers about high-risk segments, geographic and demographic differences, and the financial profile of churned customers. This project addresses that gap through reproducible segmentation and KPI analysis.

Primary objectives

- Measure the overall churn rate.
- Identify churn distribution across customer segments.
- Compare churn behavior across France, Spain, and Germany.

Secondary objectives

- Understand churn among high-value customers.
- Evaluate engagement and tenure patterns.
- Support targeted retention and marketing decisions.

3. Dataset and data quality

The linked source file contains **10,000 rows and 14 columns**. The portal brief lists the customer fields and the source file contains one additional field, Year. All records have Year = 2025, so it is retained as source metadata but not used for time-series analysis.

Validation results

Check	Result
Source rows	10,000
Final analysis rows	10,000
Duplicate customer IDs	0 rows
Missing values in core fields	0 rows
Invalid binary values	None detected
Unexpected geographies	None detected
Unexpected genders	None detected
Year coverage	2025 only

The source is analytically usable without row removal. Surname is excluded from analysis because it is not needed and may identify customers. CustomerId is used for uniqueness validation and is not displayed in the dashboard.

4. Analytical methodology

Data preparation

1. Load the raw CSV and preserve an untouched copy.
2. Convert numeric fields to numeric types and normalize text fields.
3. Validate the one-row-per-customer assumption.
4. Validate the binary fields HasCrCard, IsActiveMember, and Exited.
5. Exclude Surname from analytical outputs.
6. Create reusable segmentation fields.

Segmentation definitions

The portal names several groups but does not provide all cutoffs. The project therefore uses documented provisional definitions:

- **Age:** <30, 30-45, 46-59, 60+. The portal's written 46-60 and 60+ ranges overlap at age 60, so the implementation uses 46-59 and 60+ to make groups mutually exclusive.
- **Credit score:** Low <580, Medium 580-669, High 670+.
- **Tenure:** New 0-2, Mid-term 3-6, Long-term 7+ years.
- **Balance:** zero-balance, low-balance, and high-balance. Positive balances are split at their median of 119,839.69.
- **High-value proxy:** positive balance at or above the 75th percentile of positive balances, 139,512.29.

KPI formulas

- Overall churn rate = churned customers / all valid customers.
- Segment churn rate = churned customers in segment / all customers in segment.
- Churn contribution = churned customers in segment / all churned customers.
- Geographic risk index = segment churn rate / overall churn rate x 100.

- High-value churn ratio = churned high-value customers / all high-value customers.
- Balance exposure = sum of balances for churned high-value customers.

Churn rate and churn contribution are different measures. Every table reports counts alongside percentages so small segments are not over-interpreted.

5. Results and findings

5.1 Overall churn

The dataset contains **10,000 customers**, of whom **2,037 exited** and **7,963 remained**. The overall churn rate is **20.4%**.

5.2 Geography

Segment	Customers	Churned	Retained	Churn rate	Share of churners	Risk index
Germany	2509	814	1695	32.4%	40.0%	159
Spain	2477	413	2064	16.7%	20.3%	82
France	5014	810	4204	16.2%	39.8%	79

Germany is the highest-risk geography with a churn rate of **32.4%** and a risk index of **159**. Germany contributes **40.0%** of all churned customers. France has the largest customer population, but its churn rate is much lower than Germany's.

5.3 Age

Segment	Customers	Churned	Retained	Churn rate	Share of churners	Risk index
46-59	1585	810	775	51.1%	39.8%	251
60+	526	147	379	27.9%	7.2%	137
30-45	6248	956	5292	15.3%	46.9%	75
<30	1641	124	1517	7.6%	6.1%	37

The 46-59 group is the strongest age-related signal: **810 of 1,585 customers churned**, a rate of **51.1%**. The group contributes **39.8%** of all churners. Customers under 30 have the lowest observed churn rate.

5.4 Geography and age interaction

Geography	Age group	Customers	Churned	Churn rate	Share of churners	Risk index
Germany	46-59	492	331	67.3%	16.2%	330
France	46-59	706	321	45.5%	15.8%	223
Germany	60+	123	51	41.5%	2.5%	204
Spain	46-59	387	158	40.8%	7.8%	200
Germany	30-45	1522	385	25.3%	18.9%	124
France	60+	274	68	24.8%	3.3%	122
Spain	60+	129	28	21.7%	1.4%	107
Germany	<30	372	47	12.6%	2.3%	62
Spain	30-45	1558	195	12.5%	9.6%	61
France	30-45	3168	376	11.9%	18.5%	58
Spain	<30	403	32	7.9%	1.6%	39
France	<30	866	45	5.2%	2.2%	26

The highest interaction segment is **Germany customers aged 46-59**, with a churn rate of **67.3%** (331 churners from 492 customers). This segment should be checked for operational causes before a retention campaign is designed.

5.5 Engagement and tenure

Segment	Customers	Churned	Retained	Churn rate	Share of churners	Risk index
Inactive	4849	1302	3547	26.9%	63.9%	132
Active	5151	735	4416	14.3%	36.1%	70

Inactive customers churn at **26.9%**, compared with **14.3%** among active customers. The inactive-to-active churn-rate ratio is **1.88x**. Inactive customers account for **63.9%** of all churners.

Segment	Customers	Churned	Retained	Churn rate	Share of churners	Risk index
New (0-2)	2496	528	1968	21.2%	25.9%	104
Mid-term (3-6)	3977	821	3156	20.6%	40.3%	101
Long-term (7+)	3527	688	2839	19.5%	33.8%	96

Tenure differences are comparatively modest. New customers have a churn rate of approximately **21.2%**, while long-term customers are at approximately **19.5%**. Tenure should therefore be considered together with activity, geography, and age rather than used alone.

5.6 Demographic and credit-profile comparisons

Gender

Segment	Customers	Churned	Retained	Churn rate	Share of churners	Risk index
Female	4543	1139	3404	25.1%	55.9%	123
Male	5457	898	4559	16.5%	44.1%	81

Female customers show a higher observed churn rate than male customers in this dataset. This is a descriptive difference, not evidence that gender causes churn. Recommendations should focus on the associated customer characteristics, not demographic stereotypes.

Credit score

Segment	Customers	Churned	Retained	Churn rate	Share of churners	Risk index
Low (<580)	2362	520	1842	22.0%	25.5%	108
Medium (580-669)	3331	685	2646	20.6%	33.6%	101
High (670+)	4307	832	3475	19.3%	40.8%	95

Credit score has a smaller gradient than geography, age, activity, and product count. The low-score band is above the overall churn rate, but the difference is not sufficient by itself to explain churn.

5.7 Balance and high-value customers

Segment	Customers	Churned	Retained	Churn rate	Share of churners	Risk index
High-balance	3191	771	2420	24.2%	37.8%	119
Low-balance	3192	766	2426	24.0%	37.6%	118
Zero-balance	3617	500	3117	13.8%	24.5%	68

Zero-balance customers have lower churn than positive-balance customers. Positive balance should not automatically be interpreted as customer value because the dataset does not include fees, profitability, or product revenue.

Segment	Customers	Churned	Retained	Churn rate	Share of churners	Risk index
High-value proxy	1596	367	1229	23.0%	18.0%	113
Other customers	8404	1670	6734	19.9%	82.0%	98

There are **1,596 high-value proxy customers**. Their churn rate is **23.0%**, compared with **19.9%** for other customers. Churned high-value customers represent potential balance exposure of **58,714,368.09** in the dataset's balance units. This is not confirmed lost revenue.

5.8 Number of products

Segment	Customers	Churned	Retained	Churn rate	Share of churners	Risk index
4	60	60	0	100.0%	2.9%	491
3	266	220	46	82.7%	10.8%	406
1	5084	1409	3675	27.7%	69.2%	136
2	4590	348	4242	7.6%	17.1%	37

Product count is a strong observed discriminator. Customers with four products have a 100% churn rate, but the group contains only 60 customers. Customers with three products also show very high churn, while the two-product group has the lowest rate. These values should be validated with the mentor because product counts of three and four may represent a data or business-process issue rather than a stable causal pattern.

6. Recommendations

The recommendations below are directly connected to the observed patterns:

1. **Prioritize Germany for retention diagnostics.** Germany has the highest churn rate and risk index. Review product mix, service interactions, and inactive-customer share in that country before deploying a country-specific campaign.

2. **Create a focused 46-59 retention segment.** This age group has the highest churn across the overall dataset and within each geography. Use tailored relationship reviews rather than generic messaging.

3. **Run an inactivity re-engagement program.** Inactive customers churn at roughly 1.88 times the active-customer rate. Prioritize inactive customers in high-risk geography-age combinations.

4. **Investigate three- and four-product customers.** Their churn rates are extreme, but the groups are small. Confirm whether these product counts represent product closures, data quality issues, or genuinely unstable relationships.

5. **Protect high-balance relationships.** High-value proxy customers have slightly higher churn and meaningful balance exposure. Use balance as a prioritization signal, not as a complete customer-value measure.

6. **Track segment rates and counts together.** Retention teams should monitor churn rate, churner volume, risk index, and potential balance exposure for every campaign.

7. Dashboard design

The Streamlit application implements the portal requirements through:

- an overall churn summary;
- geography-wise churn visualization;
- age and tenure comparisons;
- credit, balance, activity, and product tables;
- a high-value customer explorer;
- filters for all major segments;
- dynamic KPI updates; and
- geography-age drill-down analysis.

The dashboard excludes surnames and does not display row-level customer records.

8. Limitations and ethical considerations

- All rows are from 2025, so the dataset does not support time-series churn analysis.
- There is no revenue, profit, transaction, complaint, service-quality, or product-name field.
- Balance exposure is a proxy, not actual revenue loss.
- The data is observational and does not establish causality.
- Small groups, especially three- and four-product customers, require caution.
- Demographic and geographic comparisons must not be converted into stereotypes or automatic adverse decisions.
- The dashboard should support human review and targeted investigation, not automatically deny service or pricing.

9. Conclusion

The most important observed churn signals are geography, age, activity status, and number of products. Germany and the 46-59 age group have the strongest segment-level risk, inactive customers account for most churners, and high-value customers show meaningful potential balance exposure. The resulting dashboard and written deliverables provide a reproducible starting point for retention analysis while clearly separating observed associations from unsupported causal or revenue claims.

Appendix: reproducibility

Run the following commands from the project root:

```
python3 scripts/build_analysis_outputs.py
python3 -m unittest discover -s tests -v
streamlit run app.py
```

The source data is preserved under data/, the prepared internal analysis data is under data/processed/, and computed tables/charts are under outputs/.